

Group Fairness in Machine learning



Conditions:

- Independence
- Sufficiency
- Separation



Independence:
The Unbiased Classifier

Independence in machine learning means fair algorithms make predictions without bias based on attributes like race, gender, or age. This ensures equal chances for all, free from inherent prejudices.



Sufficiency:
Ensuring Equitable Accuracy

Sufficiency in group fairness ensures that predicted risks are equally accurate for all groups, regardless of sensitive information. This principle guarantees that predicted outcomes are sufficient for decision-making across all groups.



Separation:
Fairness of Distinct Outcomes

Separation requires algorithms to be unbiased regarding personal information when predicting outcomes. This principle is crucial in high-stakes areas like criminal justice or healthcare, ensuring errors are evenly spread across all groups to prevent systematic disadvantage.



Usage Examples

Law Enforcement: Prevents biases that may result in excessive surveillance or targeting of specific racial or ethnic groups

Public services: Prevent demographically made systematic exclusion when allocating resources like housing assistance or welfare benefits.



Real World Examples

Equity: Ensure fair access and treatment in services and opportunities, leveling the playing field for all demographics.

Decision Quality Improvement: Elevates the accuracy and reliability of AI decisions, benefiting individuals and communities alike.

